`script1.py` — Summary

What It Does in One Sentence

For every gene, it finds **where ribosomes are concentrated along the mRNA** — expressed as a single percentage number — and does this across two experimental conditions (with and without CHX drug treatment).

Inputs

2 JSON config files (one per condition — with CHX, without CHX). Each config points to **6 HDF5 files**:

```
3 × TOM interactome replicates
3 × Translatome replicates        ┐  (× 2 conditions = 12 HDF5 files total)
```

Each HDF5 file contains, per gene:

- A list of **positions** along the CDS (codon positions)
- The **ribosome count** at each position
- Metadata: `gene_id`, `protein_id`, `cds_length`, `tag`

What Happens Step by Step (Simplified)

```
1. Load all 6 HDF5 files for a condition
         ↓
2. Keep only genes found in ALL 6 files
   (if a gene is missing from even one replicate, it's dropped)
         ↓
3. Filter out low-expression genes
   (drop genes below the median read count in the interactome replicates)
         ↓
4. For each surviving gene, in each replicate:
   → Sort positions 3′ to 5′
   → Accumulate counts until you reach 90% of total
   → Record that position as a % of CDS length  ← this is the K% number
         ↓
5. Average the K% across replicates
   + compute standard deviation
         ↓
6. Repeat steps 1–5 for the second condition (without CHX)
         ↓
7. Merge both conditions side by side on gene name → save CSV
```

The Core Number: What Is K% CDS Position?

Imagine a gene's mRNA laid out left (5′) to right (3′). Ribosomes sit at various positions along it. The script asks:

"Starting from the 3' end, how far back do I need to go to account for 90% of all ribosome counts?"

That distance, expressed as a percentage of total CDS length, is the **K% CDS position**.

- **High value (~80–100%)** → ribosomes spread across the whole mRNA → normal translation
- **Low value (~10–30%)** → ribosomes piled up near the start → stalling or early arrest

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Outputs

File	What It Is
`df_with_scatter_input.csv`	K% values for the with-CHX condition
`df_without_scatter_input.csv`	K% values for the without-CHX condition
`final_merged_output.csv`	Both conditions merged — **main output**
`run_summary.txt`	Log of everything that happened during the run

The final CSV has one row per gene, with columns for each replicate's K% value, plus the mean and standard deviation across replicates, for both conditions.